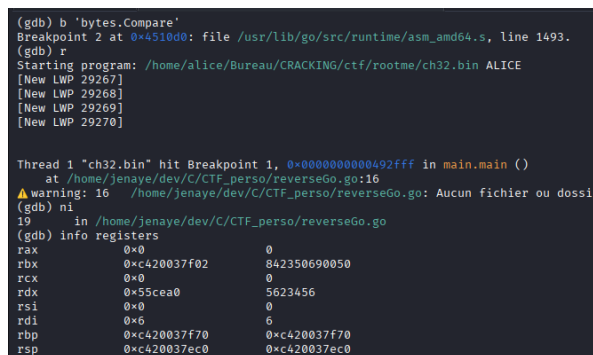
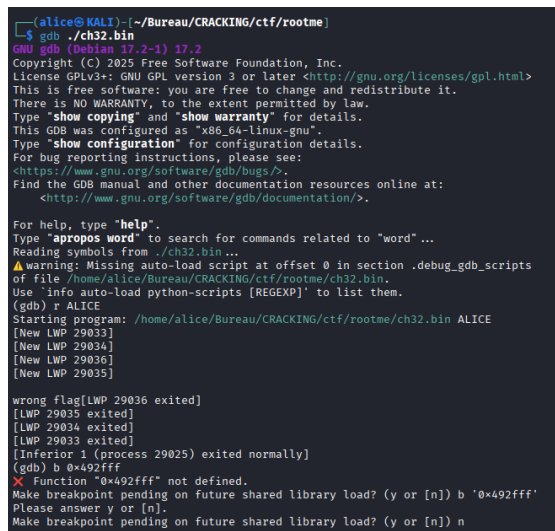
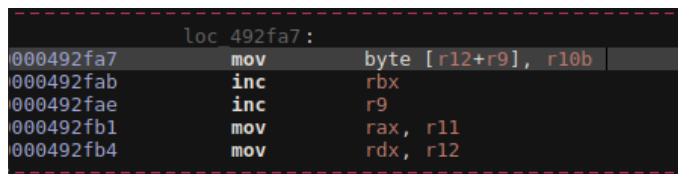
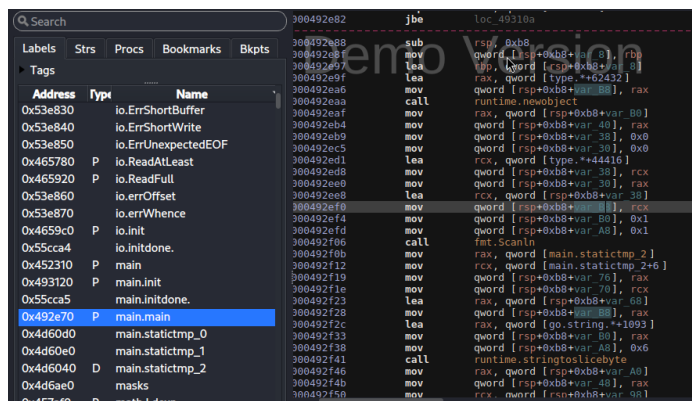
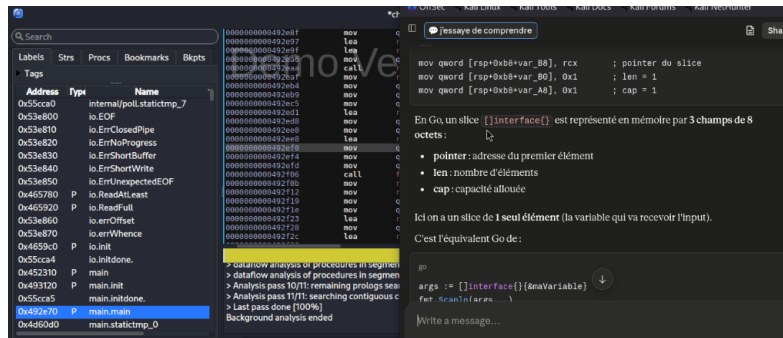


GO

dimanche 7 juin 2026 17:37

Avec hopper,

On peut voir que notre input est récupéré par le registre rcx lors de l'initialisation de la fonction
fmt.Scanln



```

r8      0xc420037f10      842350690064
r9      0x0                0
r10     0x52ea80          5433984
r11     0x206             518
r12     0x0                0
r13     0xf3              243
r14     0x33              51
r15     0x28              40
rip     0x493004          0x493004 <main.main+404>
eflags  0x246             [ PF ZF IF ]
cs      0x33              51
ss      0x2b              43
ds      0x0                0
es      0x0                0
fs      0x0                0

```

```

(gdb) x /16xb $rbx
0xc420037f02: 0x3b 0x02 0x23 0x1b 0x1b 0x0c 0x1c 0x08
0xc420037f0a: 0x28 0x1b 0x21 0x04 0x1c 0x0b 0x72 0x6f
(gdb) x /32xb $rbx
0xc420037f02: 0x3b 0x02 0x23 0x1b 0x1b 0x0c 0x1c 0x08
0xc420037f0a: 0x28 0x1b 0x21 0x04 0x1c 0x0b 0x72 0x6f
0xc420037f12: 0x6f 0x74 0x6d 0x65 0x00 0x00 0x00 0x00
0xc420037f1a: 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00
(gdb) x /16xb $rbx
0xc420037f02: 0x3b 0x02 0x23 0x1b 0x1b 0x0c 0x1c 0x08
0xc420037f0a: 0x28 0x1b 0x21 0x04 0x1c 0x0b 0x72 0x6f
(gdb) x /64xb $rbx
0xc420037f02: 0x3b 0x02 0x23 0x1b 0x1b 0x0c 0x1c 0x08
0xc420037f0a: 0x28 0x1b 0x21 0x04 0x1c 0x0b 0x72 0x6f
0xc420037f12: 0x6f 0x74 0x6d 0x65 0x00 0x00 0x00 0x00
0xc420037f1a: 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00
0xc420037f22: 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00
0xc420037f2a: 0x00 0x00 0x00 0x00 0x00 0x00 0x10 0x7f
0xc420037f32: 0x03 0x20 0xc4 0x00 0x00 0x00 0xa0 0xa0
0xc420037f3a: 0x00 0x20 0xc4 0x00 0x00 0x00 0x80 0xed
(gdb) x /s $rbx
0xc420037f02: ";\\002#\\033\\033\\f\\034\\b(\\033!\\004\\034\\vrootme"
(gdb) x/s $rbx
0xc420037f02: ";\\002#\\033\\033\\f\\034\\b(\\033!\\004\\034\\vrootme"
(gdb) x/s $rbx
0xc420037f02: ";\\002#\\033\\033\\f\\034\\b(\\033!\\004\\034\\vrootme"
(gdb) x/s $r8
0xc420037f10: "rootme"
(gdb)

```

Dans le programme `byte_Compare` :

```

warning: 1493 /usr/lib/go/src/runtime/asm_amd64.s: Aucun fichier ou dossier de ce nom
(gdb) i r rbx
rbx      0xc420037f02      842350690050
(gdb) info registers
rax      0x0                0
rbx      0xc420037f02      842350690050
rcx      0x0                0
rdx      0x55cea0          5623456
rsi      0x0                0
rdi      0x6                6
rbp      0xc420037f70      0xc420037f70
rsp      0xc420037eb8      0xc420037eb8
r8       0xc420037f10      842350690064
r9       0x0                0
r10      0x52ea80          5433984
r11      0x202             514
r12      0x0                0
r13      0xf3              243
r14      0x33              51
r15      0x28              40
rip      0x4510d0          0x4510d0 <bytes.Compare>
eflags   0x246             [ PF ZF IF ]
cs       0x33              51
ss       0x2b              43
ds       0x0                0
es       0x0                0
fs       0x0                0
gs       0x0                0
fs_base  0x53f450          5502032
gs_base  0x0                0
(gdb) i r r8
r8       0xc420037f10      842350690064
(gdb) x/s $r8
0xc420037f10: "rootme"
(gdb) x/s $r10
0x52ea80 <runtime.class_to_size>: ""
(gdb) x/s $rbx
0xc420037f02: ";\\002#\\033\\033\\f\\034\\b(\\033!\\004\\034\\vrootme"
(gdb)

```

```

(gdb) si
runtime.cmpbody () at /usr/lib/go/src/runtime/asm_amd64.s:1507
1507 in /usr/lib/go/src/runtime/asm_amd64.s
(gdb) info registers
rax      0x0                0
rbx      0xe               14
rcx      0x0                0
rdx      0x0                0
rsi      0xc420037f02      842350690050
rdi      0x55cea0          5623456
rbp      0xc420037f70      0xc420037f70
rsp      0xc420037eb8      0xc420037eb8
r8       0xc420037f10      842350690064
r9       0xc420037ef0      842350690032
r10      0x52ea80          5433984
r11      0x202             514
r12      0x0                0
r13      0xf3              243
r14      0x33              51
r15      0x28              40
rip      0x4510f0          0x4510f0 <runtime.cmpbody>
eflags   0x246             [ PF ZF IF ]
cs       0x33              51
ss       0x2b              43
ds       0x0                0
es       0x0                0
fs       0x0                0
gs       0x0                0
fs_base  0x53f450          5502032
gs_base  0x0                0
(gdb) x/s $r9
0xc420037ef0: ""\\337J"
(gdb)

```

```

mov     rdx, [rbx]
mov     rdi, [rsp+0x80+var_80]
mov     r9, [rsp+0x80+var_40] ; on met rootme dans le registre r8
xor     r9d, r9d
jmp     short loc_492FA7 ; on compare r9 (rootme) avec rsi qui devrait contenir une taille (donc on boucle ici)

loc_492FA7:
mov     [r12+r9], r10b ; CODE XREF: main.main+189j
inc     r9 ; on met notre input dans d'adresse de r12+r9
inc     r9 ; on incrémente rdx (peut être on passe au byte suivant de notre input)
mov     rax, r11 ; on incrémente r9 de 1 (peut être on passe au byte suivant de rootme)
mov     rdx, r12

loc_492FB7:
cmp     r9, rsi ; CODE XREF: main.main+135j
jge     short loc_492FF7 ; on compare r9 (rootme) avec rsi qui devrait contenir une taille (donc on boucle ici)
movzx   r10d, byte ptr [rbx] ; rdx compare une valeur après le branch
test    rdi, rdi ; on met un byte de notre input dans r10d
je      loc_493103
mov     r11, rax ; rootme dans rax
mov     rax, r9
mov     r12, rdx
cmp     rdi, 0xffffffffffffh
jz      short loc_492FF7
cgo     rdi
jmp     short loc_492FE4

```

```

ID... Stack of ... H... Lo... I... E... (gdb) x/16xb $r8
0xc420037f02: 0x55 0x55 0x55 0x55 0x55 0x55 0x55 0x55
0xc420037f0a: 0x55 0x55 0x55 0x55 0x55 0x55 0x55 0x55

```

```

00032FFF 000000000492FF: main_m (Synchronized with Hex V1
[
    lea     rbp, [rspp=0x0+var_10 away], 0
    add     rax, rax
    ; CODE XREF: main_main+24A:1
    call    __fprintf
    (gdb) next
    10 /home/jenaye/dev/CTF/perso/reversego.go
    (gdb) continue
    Continuing.
    Thread 1 'ch32.bin' hit Breakpoint 1, 0x000000000492ffff in main.
    at /home/jenaye/dev/CTF/perso/reversego.go:10
    (gdb) x/16b $rsp+0x42
    0xc420037f02: 0-b3 0-b2 0x21 0-b1b 0-b1b 0-b0c 0-b
    0xc420037f0a: 0-b26 0-b1c 0x21 0-b4 0-b1c 0-b0b 0-b

```

Et puis, si on suit la fonction 42FF on peut voir qu'on transfère l'adresse d'une variable dans rbx.

On voit bien l'opération `lea rbx, [rsp+0xb8h+var_76]` que j'ai changé en `0x42h`, car on peut soustraire cette valeur avec ce qu'on a réservé pour la variable 76 :

```

; CODE XREF: main_main+14A:J
loc_492FFF:
lea     rbx, [rsp+42h] ; rbx récupère une valeur après la boucle
mov     [rsp+0B8h+4*str], 0Bh ; on met le type de notre input dans cette adresse
mov     [rsp+0B8h+a.len], 0Bh
mov     [rsp+0B8h+1], rdx ; n
mov     [rsp+0B8h+err.data], rcx ; err
mov     [rsp+0B8h+err.data], rcx
call    bytes_Compare ; on compare notre variable avec quelque chose
mov     rcx, [rsp+0B8h+var_08]
test    rcx, rcx
jnz     short loc_4930A1
mov     [rsp+0B8h+var_18.array], 0
mov     [rsp+0B8h+var_18.len], 0
lea     rcx, RTYPE_string
mov     [rsp+0B8h+var_18.array], rcx
lea     rcx, main_statictmp_0
mov     [rsp+0B8h+var_18.len], rcx
mov     rcx, [rsp+0B8h+var_18]
mov     [rsp+0B8h+typ], rcx ; a
mov     [rsp+0B8h+str], 1
mov     [rsp+0B8h+a.len], 1
call    fmt_Print

loc_493091:
mov     rbp, [rsp+0B8h+var_18.cap]
add     rbp, 0B8h
; CODE XREF: main_main+28A:J
492FFF: main_main+loc_492FFF (Synchronized with Hex View-1)

```

Et avec dbg on peut voir ce que contient cette adresse pointer :

```
Thread 1 "ch32.bin" hit Breakpoint 1, 0x0000000000492fff in main.main ()
    at /home/jenaye/dev/C/CTF_perso/reverseGo.go:16
16      in /home/jenaye/dev/C/CTF_perso/reverseGo.go
(gdb) x/16b $rsp+0x42
0xc420037f02: 0x3b 0x02 0x23 0x1b 0x1b 0x0c 0x1c 0x08
0xc420037f0a: 0x28 0x1b 0x21 0x04 0x1c 0x0b 0x72 0x6f
(gdb)
```

Ok maintenant on peut être récupérer ses bytes et voir avec quoi elles sont comparés :

```
Thread 1 "ch32.bin" hit Breakpoint 1, 0x000000000492fff in main.main ()
    at /home/jenaye/dev/C/CTF_perso/reverseGo.go:16
16      in /home/jenaye/dev/C/CTF_perso/reverseGo.go
(gdb) x/16b $rsp+0x42
0x420037f02: 0x3b 0x02 0x23 0x1b 0x1b 0x0c 0x1c 0x08
0x420037f0a: 0x28 0x1b 0x21 0x04 0x1c 0x0b 0x72 0x6f
(gdb) x/16b $rsp+0x30
0x420037ef0: 0x60 0xdf 0x4a 0x00 0x00 0x00 0x00 0x00
0x420037ef8: 0x06 0x00 0x00 0x00 0x00 0x00 0x00 0x00
(gdb) x/s $rsp+0x30
0x420037ef0: "" "3373"
(gdb) x/1s $rsp+0x42
0x420037f02: ";!002!033!033!f!034!b!\\033!\\004!034!vrootme"
(gdb)
```

Hmm.. Ok.

Il faudrait re-xore cette valeur avec rootme qui est la clé.

Je ne comprends pas ici comment on a su que rootme ct une constante:..

```

Thread 1 "ch22.bin" hit breakpoint 5, bytes:Compare ()
at /usr/lib/gdb/src/runtime/asm_aarch64.s:1497
warning: 1497 /usr/lib/gdb/src/runtime/asm_aarch64.s: Aucun fichier ou dossier de
ce type.
(gdb) info registers
rax      0x0
rbx      0x0
rcx      0x0
rdi      0x0
rsi      0xc2003df02      0x230714626
rsd0     0x550ca00      50230456
rsp      0xc2003df00      0xc2003df70
rtp      0xc2003df00      0xc2003de68
r8        0xc2003df10      80230714640
r9        0x0
r10       0x2ea80      5433984
r11       0x206      518
r12       0x0
r13       0xf3      243
r14       0x33      51
r15       0x28      40
r16       0x4510e4      0x4510e4 <bytes.Compare+20>
rflags   [ 2f 1f ]
cs        0x33      51
ss        0x2b      43
ds        0x0      0
fs        0x0      0
gs        0x0      0
fs_base   0x3f450      5502032
gs_base   0x0
(gdb) x /s $r15
0x28: <error: Cannot access memory at 0x28>
(gdb) x /s $r8
0xc2003df10: "rootme"
(gdb) x/s $rsp+0x38
0xc2003de68: "\337?"

```

On peut voir ici que le registre r8 contient bien rootme à chaque fois.

La valeur passé à r9 est celle-ci : \337J

```
0x000000000049310f <+b/1>: jmp 0x492e70 <main.main>
```

```
(gdb) $rsp+0x42
X Undefined command: "$rsp+0x42". Try "help".
(gdb) info registers rsp
rsp      0xc420037ec0      0xc420037ec0
(gdb) x/16x 0xc420037ec0
0xc420037ec0: 0xc0  0x46  0x4a  0x00  0x00  0x00  0x00  0x00
0xc420037ec8: 0x00  0x00  0x00  0x00  0x00  0x00  0x00  0x00
(gdb) x /s $rsp
0xc420037ec0: "\300FJ"
(gdb) x/s $rsp+0x42
quit
0xc420037f02: ";!002#\033\033!\f\034\b(\033!\004\034\vroutme"
(gdb) x/16bx $rsp+0x42
0xc420037f02: 0x3b  0x02  0x23  0x1b  0x1b  0x0c  0x1c  0x08
0xc420037f0a: 0x28  0x1b  0x21  0x04  0x1c  0x0b  0x72  0x6f
(gdb) x/32bx $rsp+0x42
0xc420037f02: 0x3b  0x02  0x23  0x1b  0x1b  0x0c  0x1c  0x08
0xc420037f0a: 0x28  0x1b  0x21  0x04  0x1c  0x0b  0x72  0x6f
0xc420037f12: 0x6f  0x74  0x6d  0x65  0x00  0x00  0x00  0x00
0xc420037f1a: 0x00  0x00  0x00  0x00  0x00  0x00  0x00  0x00
(gdb)
```

End of assembler dump.

```
(gdb) $rsp+0x42
```

Undefined command: "\$rsp+0x42". Try "help".

```
(gdb) info registers rsp
```

```
rsp      0xc420037ec0      0xc420037ec0
```

```
(gdb) x/16x 0xc420037ec0
```

```
0xc420037ec0: 0xc0  0x46  0x4a  0x00  0x00  0x00  0x00  0x00
```

```
0xc420037ec8: 0x00  0x00  0x00  0x00  0x00  0x00  0x00  0x00
```

```
(gdb) x /s $rsp
```

```
0xc420037ec0: "\300FJ"
```

```
(gdb) x/s $rsp+0x42
```

```
quit
```

```
0xc420037f02: ";!002#\033\033!\f\034\b(\033!\004\034\vroutme"
```

```
(gdb) x/16bx $rsp+0x42
```

```
0xc420037f02: 0x3b  0x02  0x23  0x1b  0x1b  0x0c  0x1c  0x08
```

```
0xc420037f0a: 0x28  0x1b  0x21  0x04  0x1c  0x0b  0x72  0x6f
```

```
(gdb) x/32bx $rsp+0x42
```

```
0xc420037f02: 0x3b  0x02  0x23  0x1b  0x1b  0x0c  0x1c  0x08
```

```
0xc420037f0a: 0x28  0x1b  0x21  0x04  0x1c  0x0b  0x72  0x6f
```

```
0xc420037f12: 0x6f  0x74  0x6d  0x65  0x00  0x00  0x00  0x00
```

```
0xc420037f1a: 0x00  0x00  0x00  0x00  0x00  0x00  0x00  0x00
```

On doit xoré cette valeur avec rootme.

```
char key [] = "rootme";
```

```
Unsigned char rbx [] = { 0x3b, 0x02, 0x23, 0x1b, 0x1b, 0x0c, 0x1c, 0x08, 0x28, 0x1b, 0x21,
0x04, 0x1c, 0x0b, 0x72, 0x6f, 0x6f, 0x74, 0x6d, 0x65, 0x00, 0x00 };
```

```
int len = sizeof(rbx);
```

```
For(int i = 0; i < sizeof(rbx); i++){
Rbx[i] = rbx[i] ^ key[i%len];
```

```
}
```

```
For(int i = 0; i < sizeof(rbx); i++){
Printf("Key : %s"(char*)rbx);
}
```

Ce qui nous donne le programme suivant :

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
```

```
int main () {
```

```
char key [] = "rootme";
```

```
unsigned char rbx [] = { 0x3b, 0x02, 0x23, 0x1b, 0x1b, 0x0c, 0x1c, 0x08, 0x28, 0x1b, 0x21,
0x04, 0x1c, 0x0b, 0x72, 0x6f, 0x6f, 0x74, 0x6d, 0x65, 0x00, 0x00 };
```

```
int KeyLen = strlen("rootme");
```

```
int RbxLen = sizeof(rbx);
```

```
for(int i = 0; i < RbxLen ; i++){
rbx[i] = rbx[i] ^ key[i % KeyLen];
}
```

```
printf("Key : %s\n", rbx);
```

```
return 0;
```

```
}
```

Et on a le résultat suivant :

```
alice@KALI:~/CRACKING/ctf/rootme/go$ gcc key.c -o key
```

```
(alice@KALI)-[~/CRACKING/ctf/rootme/go]
$ ./key
Key : ImLovingGoLand
t
```

ImLovingGoLand

